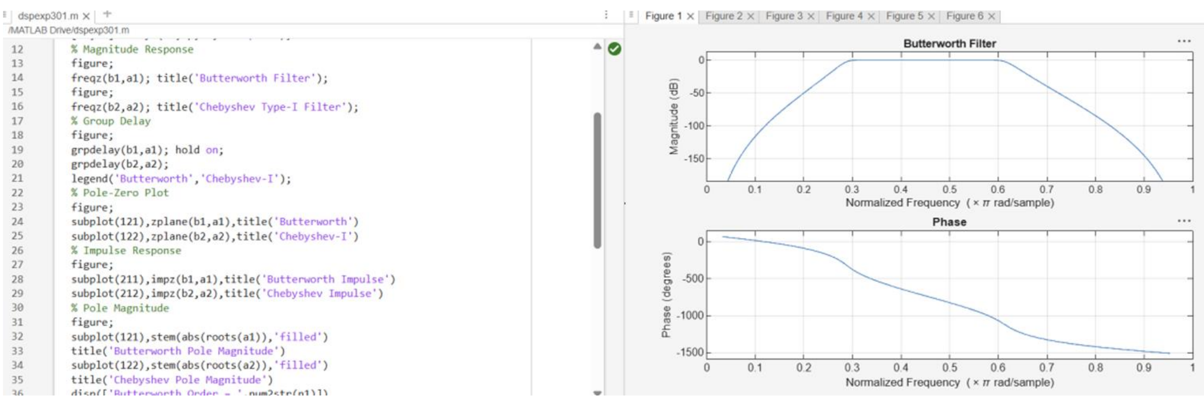
3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

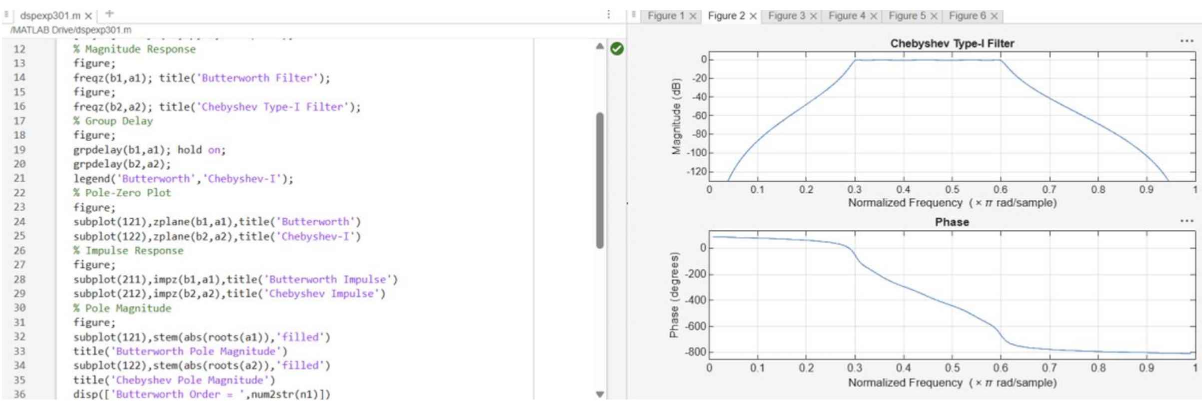
Program :

```
clc; clear; close all;
% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);
% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');
% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');
% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');
figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');
% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');
% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')
% Impulse Response
```

```
figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')
% Pole Magnitude
figure;
subplot(121),stem(abs(roots(a1)),'filled')
title('Butterworth Pole Magnitude')
subplot(122),stem(abs(roots(a2)),'filled')
title('Chebyshev Pole Magnitude')
disp(['Butterworth Order = ',num2str(n1)])
disp(['Chebyshev Order = ',num2str(n2)])
```

Output :





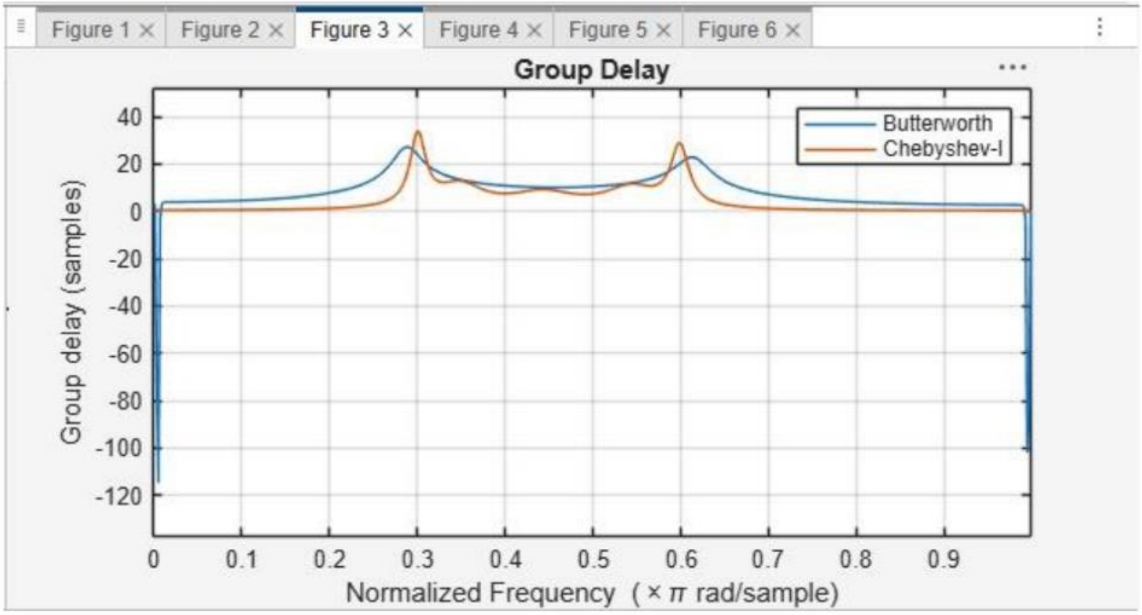


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
Group Delay
Butterworth
Chebyshev-I
Group delay (samples)
40
20
0
-20
-40
-60
-80
-100
-120
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Normalized Frequency ($\times\pi$ rad/sample)

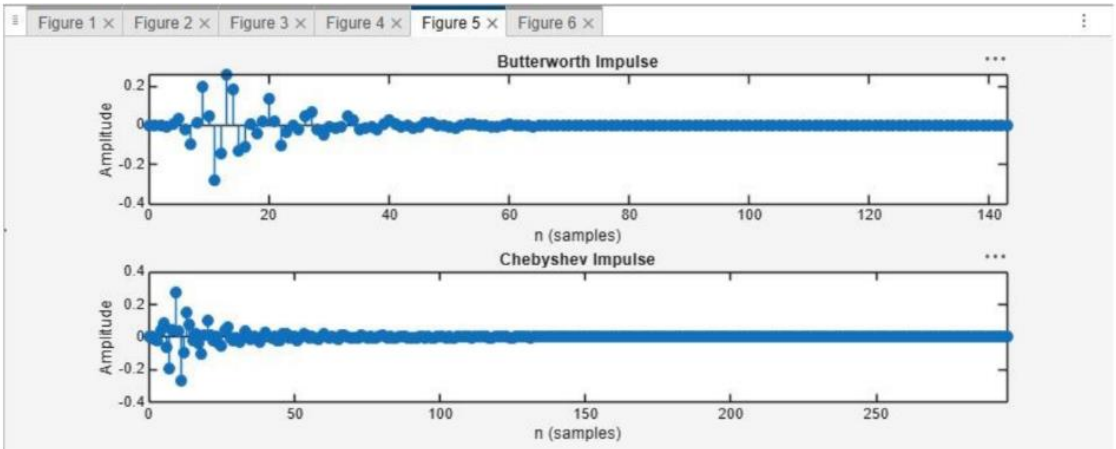


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
Butterworth Impulse
Amplitude
0.2
0
-0.2
-0.4
0
20
40
60
80
100
120
140
n (samples)
Chebyshev Impulse
Amplitude
0.4
0.2
0
-0.2
-0.4
0
50
100
150
200
250
n (samples)

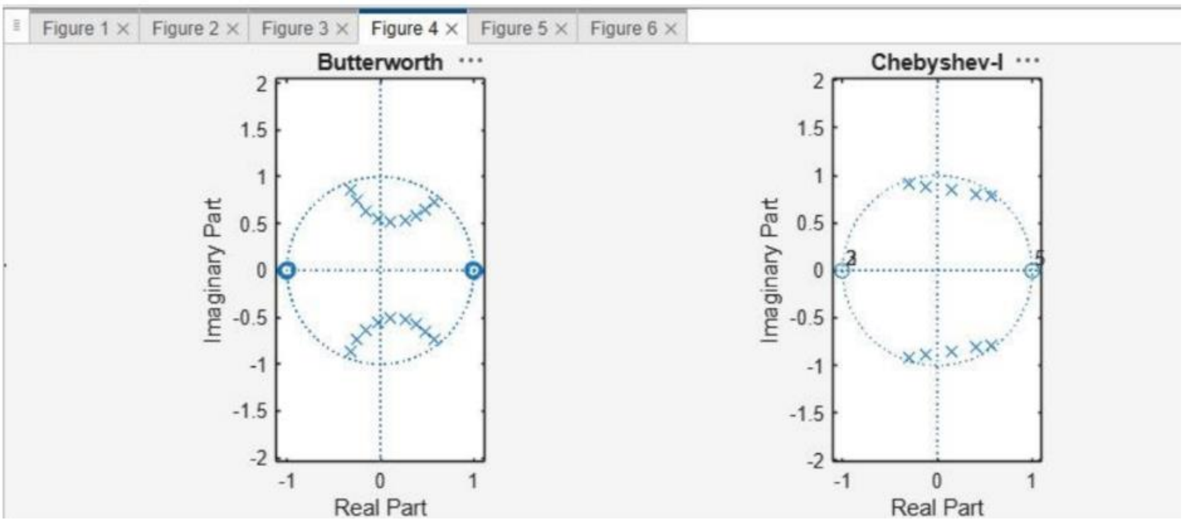


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
Butterworth
Imaginary Part
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-1
0
1
Real Part
Chebyshev-I
Imaginary Part
3
5
Real Part

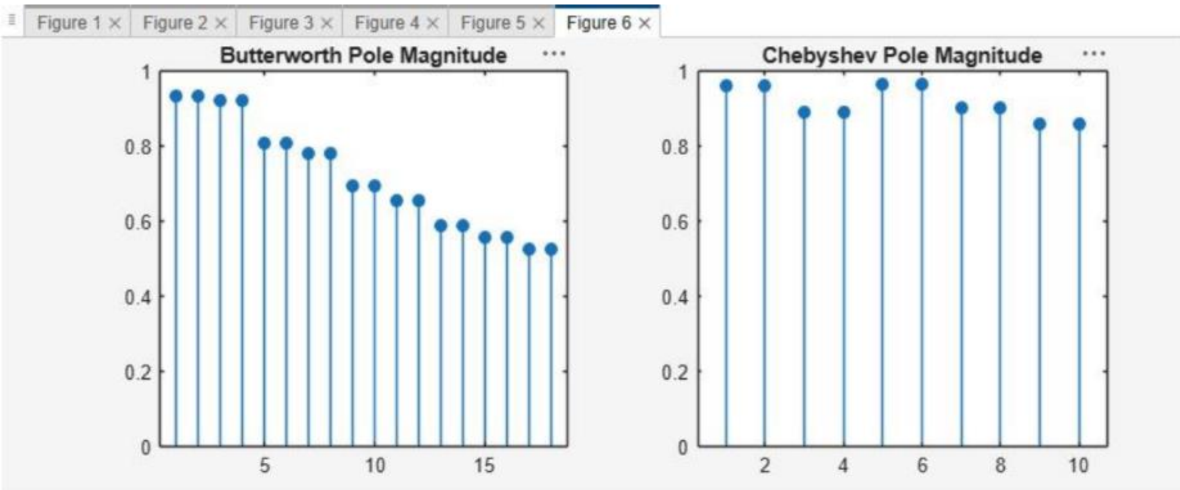


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
Butterworth Pole Magnitude
Chebyshev Pole Magnitude
1
0.8
0.6
0.4
0.2
0
5
10
15
2
4
6
8
10